

Transformer Models in Large Language Models (LLMs)

Before Transformer models, language systems were mainly based on **RNNs and LSTMs**. These models processed text **word by word**, which caused:

- Slow training
- Difficulty in handling **long sentences**
- Loss of context in large documents

This limitation made it hard to build **scalable and intelligent language systems**.

Transformer works:

Encoder (Understanding Part)

Reads the input sentence

Understands meaning and context

Used in models like BERT

Decoder (Writing Part)

Generates new text

Predicts next word step by step

Used in models like GPT

To solve this, researchers introduced the **Transformer architecture**, which is now used in **Large Language Models (LLMs)** like **ChatGPT**.

The key idea was to **replace sequential processing with attention-based processing**, allowing the model to analyze all words in a sentence at the same time.

The Transformer uses a mechanism called **self-attention**.

How it works in practice:

- Input text is broken into tokens
- Tokens are converted into embeddings
- Self-attention assigns **importance scores** to each word
- The model understands context by focusing on relevant words

- Example:
- *“The patient did not take the medicine because it was expired.”*
The model correctly links **“it”** to **“medicine”**, not “patient”.

Implementation in LLMs

In real-world LLMs:

- Multiple **attention heads** capture different relationships
- **Positional encoding** preserves word order
- **Decoder-only Transformers** (like GPT) generate text token by token

The model is trained using **next-token prediction** on massive datasets.

Results

After adopting Transformers:

- LLMs achieved **state-of-the-art accuracy**
- Training became **faster and scalable**
- Models could handle **long context**
- Applications expanded to chatbots, translation, summarization, and coding

This shows that **Transformer models solved the core limitations of earlier language models.**

By using **self-attention**, Transformers enabled the creation of powerful LLMs that understand context and generate human-like text.

